Life Protection · Public Health

JLU-SPH

2026 Modular Clay · Life Science Model Creation — Kindergarten Science Workshop Activity Plan



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1. Basic Information of the Activity

- Activity Name: Modular Clay - Life Sciences Model Creation: Kindergarten Science Workshop Activity
- Activity Theme: Building Life with Little Hands: Modular Construction of Scientific Models
- Activity Type: ☑Science Education and Popularization ☐Public Welfare Service ☐Cultural Promotion ☐Community Service ☐Other:
- Organizer: JLU-SPH Team

2. Objectives of the Activity

1. Core Objective: Using creative clay modeling as a vehicle, guide young children to recognize basic biological structures, complete the manual creation of bacteriophage, antibody (Antibody, Ab), and functional enzyme models, establish foundational cognition in life sciences and awareness of hygiene and health protection, and enable children to understand the scientific concept of "life is like building blocks" through hands-on construction.
2. Specific Objectives:

○ Knowledge/Cognitive Level: Enable children to understand the correspondence between colors and functional modules, recognize the structural logic of biological components, and gain a basic understanding of microorganisms and the body's defense mechanisms.

○ Behavioral/Skill Level: Enable children to develop manual skills such as pinching, kneading, assembling, and connecting, complete the construction of scientific models, and foster modular thinking and structural cognition.

○ Emotional/Attitudinal Level: Cultivate scientific creativity, hands-on skills, and spatial perception in children, and establish an awareness of hygiene, health, and proactive protection in daily life.

3. Expected Outcomes:

○ Number of Participants Reached: Approximately 40–50 individuals.

○ Target Satisfaction Rate: Above 90%

○ Achieve comprehensive coverage in popularizing foundational life sciences and health protection for young children

3. Target Audience

- Core Audience: Preschool children in kindergarten (ages 3–5)
- Audience Characteristics: In the critical period for developing fine motor skills, fond of colorful creation and hands-on construction, highly curious about shapes and colors but relatively lacking in self-protection awareness, and thus are the key population for this health science popularization; adult guidance and accompaniment is required for completing fine operations.

4. Activity Time and Venue

- Activity Date: April 27, 2026
- Total Activity Duration: Approximately 40 minutes
- Activity Venue: Kindergarten Children's Handicraft Room

5. Activity Content and Workflow

[Overview of Activity Process]

Stage	Time	Session Name	Specific Content	Person in Charge	Materials Required/ Photography Focus
Stage	**10 min**	Venue Setup and Participant Gathering	Arrange the venue (table and chair placement, equipment testing); designate creation and display areas; cover tables with waterproof mats and arrange materials	General Coordinator, Logistics Staff	Tables, chairs, operation mats, clay, tools, photography equipment, banners
Main Stage	**10 min**	Session 1: Science Enlightenment Explanation	With the theme “Life is like building blocks,” use child-friendly language to explain the functions of color modules, display completed models, and explain basic biological structures and health protection knowledge.	Lecturer	PPT (PowerPoint presentation), model samples
Main Stage	**20 min**	Session 2: Modular Clay Creation	Children independently select the model type (phage/bacteriophage, antibody, or Cas12a), and perform kneading, molding, and assembling operations according to color-coded modules. The teaching assistant provides one-on-one guidance to correct assembly techniques and simultaneously explains structural and functional features.	All Teaching Assistants, Lecturer	Clay, tools, operation mats, reference images
Conclusion Stage	**10 min**	Venue Restoration	Clean the tables, collect and recycle materials, and return the venue to its original state.	Teaching Assistant	Cleaning supplies, storage boxes

[Detailed Description of Each Session]

- Session 1: Science Enlightenment Explanation

- **Core Content:** Open with the theme “Life is like building blocks,” explaining in age-appropriate language that “different colors of clay are small parts of life, and when assembled together, they become amazing biological models.” Display the finished models (such as bacteriophage and antibody) and use familiar analogies to describe their functions (e.g., “The bacteriophage is a little guardian that destroys harmful bacteria,” “Antibodies are protective shields inside the body”). Present basic health protection knowledge.
- **Key Phrases:** “Children, different colors of clay are small parts of life. Put them together and they become amazing biological models!” “Bacteriophages are like little guardians in our bodies that specifically eliminate harmful bacteria!”

- **Session 2: Modular Clay Creation**

- **Core content:** Children independently select the model type (phage/bacteriophage, antibody, or Cas12a), and the teaching assistant distributes corresponding colored clay modules. Following color-coded instructions such as "red for the shell, blue for the core, green for the connector," participants perform kneading, molding, assembling, and connecting operations. The teaching assistant provides comprehensive support, corrects assembly techniques, and explains the structural and functional significance of each module throughout the process.
- **Key Phrases:** “Use red clay for the shell, blue for the core; gently knead, carefully attach, and soon our little model will be finished!” “Look, when you put the shell and core together, this becomes the head of a bacteriophage!”

VI. Personnel Allocation and Division of Responsibilities

- **General Supervisor (1 person):** Responsible for overall coordination, liaising with the school, overseeing the workflow and timeline, and handling unexpected issues. Strong communication skills and the ability to manage the entire activity are required.
- **Main Lecturer (1 person):** Responsible for scientific popularization and model demonstration in Session 1. Should possess basic biological knowledge (biology), and be skilled at communicating in a child-friendly and engaging manner.
- **On-site Teaching Assistants (3 persons):** Responsible for registration, distribution of materials, guiding groups of children in clay modeling, and maintaining order. Patience and attention to detail are necessary, as well as proficiency in caring for young children.
- **Photography and Logistics Support (2 persons):** Responsible for material management, venue cleaning, and emergency support.

VII. List of Materials

- Equipment:

- ○ Laptop computer, projector
- ○ Camera/high-resolution mobile phone
- ● Teaching Materials:
- ○ Popular Science PPT (Popular Science PowerPoint Presentation)
- ○ Clay models
- ○ Activity feedback questionnaires
- ● Logistics Supplies:
- ○ Non-toxic, food-grade clay
- ○ Handicraft operation mats
- ○ Standard first aid kit
- ○ Cleaning wet wipes, garbage bags

VIII. Risk Contingency Plan

- Accidental clay ingestion/skin allergy: Use food-grade, non-toxic clay . Before the activity, teachers and parents must confirm that there are no known allergies in children. In case of accidental ingestion or allergic reaction, immediately cease the activity, manage with the first aid kit, contact parents, and seek medical attention promptly.
- Temporary unavailability of venue: Confirm an alternative venue (e.g., backup classroom) with the kindergarten in advance. The general supervisor is responsible for early communication and confirmation.
- Insufficient clay/tools: Maintain backup logistical supplies (an extra 2–3 sets of clay and tools) to ensure all participants have adequate materials.

IX. Effectiveness Evaluation

- Process evaluation: By on-site observation of children's participation, duration of concentration, and enthusiasm in Q&A interaction, record children's understanding of the color–function module correspondence and core concept that "life is like building blocks."
- Satisfaction evaluation: Distribute simple questionnaires to the supervising teachers to collect feedback on activity content, format, and organization. Collect children's participation experience orally.